

Dynamic Macroeconomics II

Felipe Meza

Fall 2019

Fiscal Policies in the Growth Model

Second Part

Version revised September 4, 2019

- Let z_t be the vector

$$z_t = [g_t \ \tau_{it} \ \tau_{kt} \ \tau_{ct}]'.$$

- Call the difference equation

$$H(k_t, k_{t+1}, k_{t+2}, z_t, z_{t+1}) = 0.$$

- Suppose that

$$\lim_{t \rightarrow \infty} z_t = \bar{z}.$$

Calculating the steady state

- Calculate steady-state of the model by solving

$$H(\bar{k}, \bar{k}, \bar{k}; \bar{z}, \bar{z}) = 0.$$

- Let $\bar{k}$ be the steady-state value of capital. From Euler equation:

$$1 = \beta \left[\frac{1 - \tau_k}{1 - \tau_i} f'(\bar{k}) + (1 - \delta) \right].$$

- If $f(k) = Ak^\alpha$, then

$$\bar{k} = \left(\frac{1 - \tau_k}{1 - \tau_i} \frac{\alpha A}{\frac{1}{\beta} - 1 + \delta} \right)^{\frac{1}{1-\alpha}}.$$

Effects of parameters and taxes:

- Higher productivity yields higher capital.
- Higher depreciation reduces capital.
- Higher tax on payment to capital reduces capital.
- Higher subsidy to investment increases capital.

Numerical solution method for nonlinear difference equation: Shooting algorithm

- Goal: find a path for capital.
- Use Euler equation, feasibility and prices.
- Method finds initial value for consumption c_0 such that Euler equation implies $k_s \approx \bar{k}$, for a large S .
- Shooting requires $S > T$, where T is a period in which exogenous variables are constant from that period onwards.

Steps:

- 1 Find $\bar{k}$. k_0 is given.
- 2 Choose S , T and guess for c_0 . Compute k_1 using feasibility:

$$f(k_0) = c_0 + g_0 + k_1 - (1 - \delta)k_0$$

- 3 Use Euler equation to find c_1 :

$$u'(c_0) = \beta u'(c_1) \frac{1 + \tau_{c0}}{1 + \tau_{c1}} \left[\frac{1 - \tau_{k1}}{1 - \tau_{i1}} f'(k_1) + \frac{1 - \tau_{i1}}{1 - \tau_{i0}} (1 - \delta) \right]$$

and feasibility to get k_2 .

- 4 Repeat Step 3 to generate sequence $\{k_0, k_1, \dots, k_S\}$.
- 5 Compute $k_S - \bar{k}$.
- 6 If $k_S < \bar{k}$, reduce c_0 and recompute sequence for capital.
Intuition: less consumption implies more investment.
- 7 If $k_S > \bar{k}$, do the opposite.

How to update guess for c_0 ?

- Define two values for consumption: a “low” (c^ℓ) and a “high” (c^h) such that:

$$c^\ell = 0$$

$$c^h = f(k_0) + (1 - \delta)k_0$$

- The high value is equivalent to choosing $k_1 = 0$.
- Make an initial guess $c_0 = \frac{1}{2}(c^\ell + c^h)$.
- If $k_S > k$, make $c^{\ell'} = c_0$.
- We are making **new** minimum level of consumption higher.
- **New** guess $c'_0 = \frac{1}{2}(c^{\ell'} + c^h)$ will have a higher value.
- If $k_S < k$, let $c^{h'} = c_0$ and guess $c'_0 = \frac{1}{2}(c^\ell + c^{h'})$.

- Once we have computed optimal path for capital, easy to construct rest of equilibrium allocation:

$$c_t = f(k_t) + (1 - \delta)k_t - k_{t+1} - g_t$$

$$q_t = \frac{\beta^t u'(c_t)}{1 + \tau_{ct}}$$

$$r_t = q_t f'(k_t)$$

$$w_t = q_t (f(k_t) - k_t f'(k_t)).$$

(Expression for q_t had the multiplier on the intertemporal budget constraint. We can calculate it by setting $q_0 = 1$. We will see it in next topic)

We can also calculate gross real after-tax interest rate between t and $t + 1$:

$$u'(c_t) = \beta u'(c_{t+1}) R_{t+1}$$

$$R_{t+1} = \frac{1 + \tau_{ct}}{1 + \tau_{ct+1}} \left[\frac{1 - \tau_{it+1}}{1 - \tau_{it}} (1 - \delta) + \frac{1 - \tau_{kt+1}}{1 - \tau_{it}} f'(k_{t+1}) \right]$$

(An important) example with constant relative risk aversion: With

$$u(c_t) = \frac{c_t^{1-\sigma} - 1}{1-\sigma}, \sigma > 0, \neq 1$$

then (**check**)

$$\ln \left(\frac{c_{t+1}}{c_t} \right) = \frac{1}{\sigma} \ln \beta + \frac{1}{\sigma} \ln R_{t+1}.$$

With this utility, **intertemporal elasticity of substitution (IES)** is $\frac{1}{\sigma}$. If we calculate derivative of log of consumption factor with respect to the real interest rate we get $\frac{1}{\sigma}$. That is the definition of IES.

- Useful method when solving nonlinear dynamic model with certainty and single endogenous state variable (e.g. capital).
- Advantages: fast. (Much) More precise than linear methods.
- Disadvantages: cannot deal with uncertainty (many paths).
Difficult to work with more than one state variable.
- To better understand how shooting algorithm works, let us see continuous-time dynamics of neoclassical model
- For a reference, see Barro and Sala-i-Martin (1995).

Continuous-time dynamics

- All variables are a function of time.
- Maximize utility:

$$\int_0^{\infty} \frac{c^{1-\sigma} - 1}{1-\sigma} e^{-\rho t} dt$$

*has the
undescent*

where $e^{-\rho t}$ is the discount factor and ρ is the discount rate.

- Notation:

$$\dot{k} = \frac{\partial k}{\partial t}$$

- Feasibility condition:

$$y = c + \dot{k} + \delta k \quad \begin{cases} K_{t+1} = K_t(1-\delta) + i_t \\ (K_{t+1} - K_t) + \delta K_t = i_t \end{cases}$$

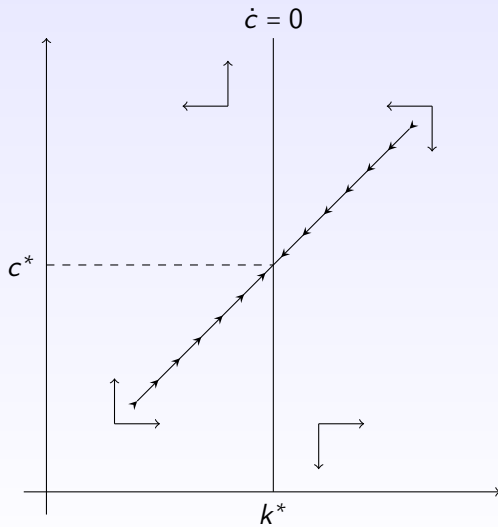
- Production function: $f(k) = k^\alpha$; inelastic labor supply = 1.
- Euler equation (you do **not** need to do the Hamiltonian):

Can't Euler Lagrange ???

$$\frac{\dot{c}}{c} = \frac{1}{\sigma} (f_k - \delta - \rho)$$

- Let us use phase diagram to illustrate dynamics of the economy.
- Model has saddle point stability.
- Shooting method tries to find the stable arm.
- If a wrong guess $c(0)$ is made, no equilibrium is reached.

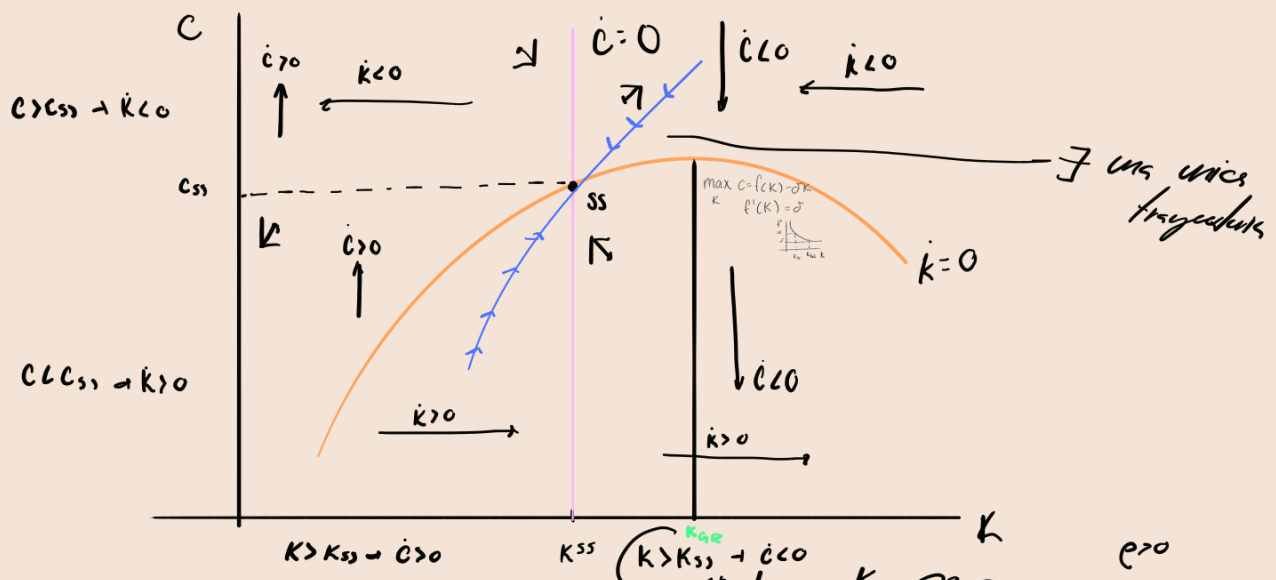
Phase Diagram



Explaining phase diagram on the blackboard

Take notes of:

- 
- Where does phase diagram come from.
 - Where do dynamics come from.
 - How to calculate steady state capital, and consumption.
 - What happens if economy starts away from stable arm.
 - What is the Golden Rule capital.
 - Why it is larger than the steady state capital.



$$\left\{ \begin{array}{l} \dot{c}/c = 1/\sigma (f'(k) - \delta - \rho) , \quad \dot{c} = 0 \Rightarrow f'(k) = \delta + \rho \\ \dot{k} = f(k) - c - \delta k , \quad c = f(k) - \delta k \end{array} \right.$$